EXPERIMENT – 1

Aim:-

Introduction to digital electronics lab- nomenclature of digital ICs, specifications, study of the data sheet, concept of Vcc and ground, verification of the truth tables of logic gates using TTL ICs.

Apparatus Required:-

Digital lab kit, single strand wires, breadboard, TTL IC’s

Gates	IC NO.
AND	7408
OR	7432
NOT	7404

Theory:-

Logic gates are idealized or physical devices implementing a Boolean function, which it performs a logical operation on one or more logical inputs and produce a single output. Depending on the context, the term may refer to an ideal logic gate, one that has for instance zero rise time and unlimited fan out or it may refer to a non-ideal physical device.

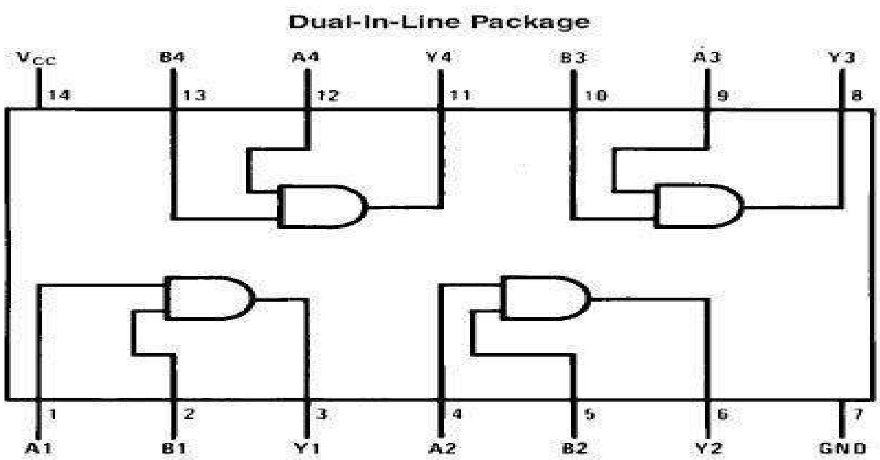
The main hierarchy is as follows:-

1. Basic Gates
2. Universal Gates
3. Advanced Gates

Basic Gates

1. **AND gate: -** Function of AND gate is to give the output true when both the inputs are true. In all the other remaining cases output becomes false. Following table justifies the statement-

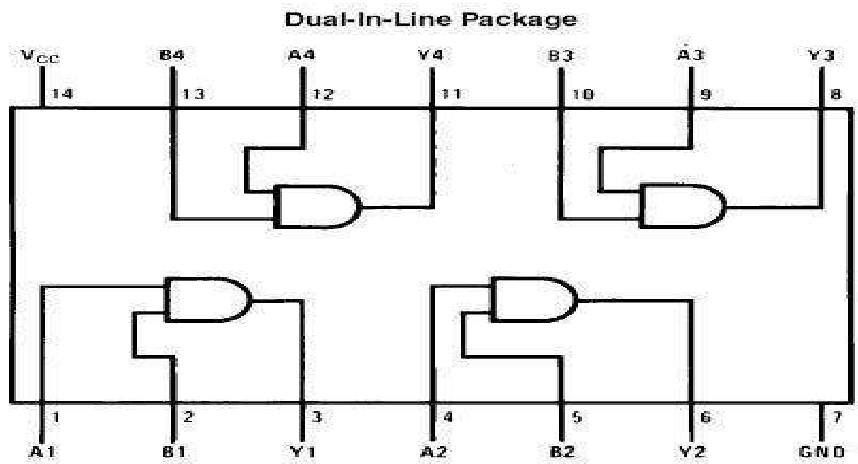
Input A	**Input B**	**Output**
1	1	1
1	0	0
0	1	0
0	0	0



IC 7408

2. **OR gate: -** Function of OR gate is to give output true when one of the either inputs are true .In the remaining case output becomes false. Following table justify the statement:-

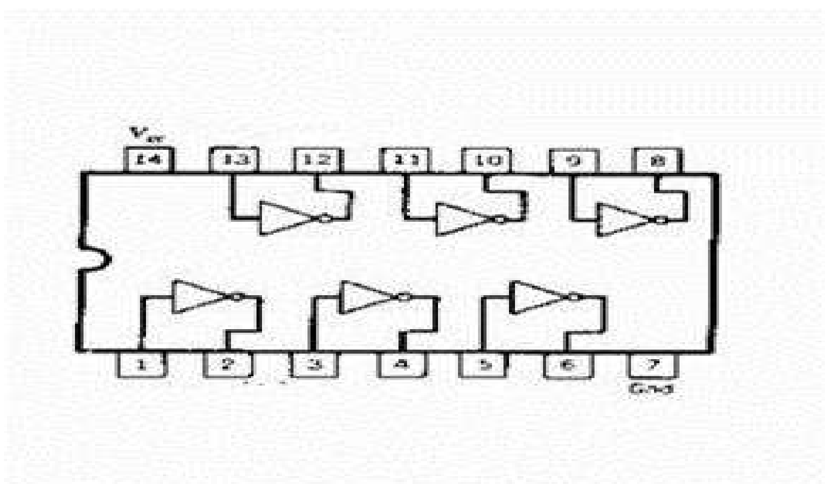
Input A	Input B	Output
0	0	0
0	1	1
1	0	1
1	1	1



IC 7432

3. **NOT gate: -** Function of NOR gate is to reverse the nature of the input .It converts true input to false and vice versa. Following table justifies the statement

Input	Output
1	0
0	1



Experiment with all the ICs and compare the result with the truth table